



Advanced Terraform on AWS

Lab 1 – Churn

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Expected Duration

This lab should take 45–60 minutes to complete.

Teaching Points – What This Lab Is Designed to Teach

This lab is deliberately not about building a sophisticated AWS network security design. It is about understanding how Terraform state, resource identity, and input data structures interact over time.

Across the phases in this lab, you will create and modify security groups inside a simple VPC. The infrastructure itself is straightforward. The learning value comes from observing how Terraform reacts as the variable structure becomes richer and the resource addressing model changes.

By the end of the lab, you should be able to clearly explain:

- Why Terraform sometimes proposes large numbers of changes even when your intent feels minimal
- Why simple ordered lists are fragile foundations for long-lived infrastructure
- How count ties resource identity to ordinal position in state
- How for_each changes resource addressing from positional indexes to stable keys
- Why map with for_each is a more stable design than list-based iteration

This lab progresses intentionally:

- Early phases create instability so you can observe it.
- Middle phases explain why it happens.
- Final phases remove the causes, not just the symptoms.

The goal is not simply to reach a clean end state, but to understand why that end state is stable and why the earlier designs were not.

Before you begin

1. Navigate to the working directory **c:\qatfadvaws\lab1\guided** which contains main.tf, variables.tf, outputs.tf, and terraform.tfvars.
2. Open a terminal in this directory.
3. This will be your working directory and root module for the lab. Make all changes in this directory only. Directory **c:\qatfadvaws\lab1\original** contains a copy of the files used in this lab and are for reference/reset purposes if needed.

Phase 1 – Initial deployment

4. Review main.tf, variables.tf, outputs.tf, and terraform.tfvars...

This configuration creates a VPC, a subnet, and a set of AWS security groups. At this stage, the security groups are driven from a simple list of strings and are created using count.

As shown in terraform.tfvars, there are currently two elements defined in the security_groups variable:

```
web
app
```

Terraform will therefore create two security group resources using the count index for addressing.

Expected state addresses:

```
aws_security_group.demo[0]
aws_security_group.demo[1]
```

Expected security group names:

```
secgp-0
secgp-1
```

Expected Name tags:

```
secgp-web
secgp-app
```

5. Deploy the resources:

```
terraform init
terraform apply --auto-approve
```

6. Verify the resources that are created in state:

terraform state list

You should see entries similar to the following:

```
aws_vpc.main
aws_subnet.demo
aws_security_group.demo[0]
aws_security_group.demo[1]
```

terraform state show aws_security_group.demo[0]

Observe carefully what Terraform is actually tracking...

```
# aws_security_group.demo[0]:
resource "aws_security_group" "demo" {
  arn = "arn:aws:ec2"
```

The state resource address is `aws_security_group.demo[0]`. That means Terraform is currently identifying the resource by list position, not by its logical meaning.

Notice that the security group name is derived from the list index (0) and the Name tag is derived from the value at that index position (web)...

```
name           = "secgp-0"
name_prefix    = null
owner_id       = "503561439433"
revoke_rules_on_delete = false
tags           = {
  "Name" = "secgp-web"
  "Role" = "web"
```

7. Verify the new resources in the AWS portal ...

Security Groups (4) Info			
Find security groups by attribute or tag			
☐	Name ▼	Security group ID ▼	Security group name
☐	secgp-web	sg-093819adde60bd368	secgp-0
☐	secgp-app	sg-05e98f5b894dfed98	secgp-1

Important: In AWS, the 'Name' of a security group is an optional tag and is mutable, whereas the 'Security group name' is mandatory and immutable.

Phase 2 – Configuration churn due to mutable attribute change

Ordinal stability

8. In **terraform.tfvars**, comment line 10 and uncomment line 11 so that **db** is added at the end of the list.
9. Plan the changes:

terraform plan

The plan should show 1 to add, 0 to change, 0 to destroy.

The new resource will be `aws_security_group.demo[2]`. This will have no effect on the existing resources `[0]` and `[1]`, because their positions in the list have not changed.

10. Apply the changes:

terraform apply --auto-approve

11. Verify the new resource in the AWS portal ...

Security Groups (5) Info			
Find security groups by attribute or tag			
☐	Name	Security group ID	Security group name
☐	secgp-web	sg-093819adde60bd368	secgp-0
☐	secgp-db	sg-051766b8238f5703a	secgp-2
☐	secgp-app	sg-05e98f5b894dfed98	secgp-1

When using count, adding items to the end of a list usually produces stable behaviour. Terraform adds the new resource and leaves the existing indexed resources untouched.

Ordinal instability

12. In **terraform.tfvars**, comment line 11 and uncomment line 12 so that **admin** is inserted at the start of the list.

13. Before running Terraform, stop and consider what you expect to happen. You have added one new security group name, but you have inserted it at the start rather than the end.

14. Plan the changes:

terraform plan

Review the plan carefully, it should show 1 to add, 3 to change, 0 to destroy.

Terraform is still using count, so the resources are still addressed as `[0]`, `[1]`, `[2]` and so on. However, the values attached to those positions have shifted.

That means:

`demo[0]` (secgp-0) Name **secgp-web** changes to **secgp-admin**

`demo[1]` (secgp-1) Name **secgp-app** changes to **secgp-web**

`demo[2]` (secgp-2) Name **secgp-db** changes to **secgp-app**

`demo[3]` (secgp-3) is created with Name **secgp-db**

15. Apply the changes:

terraform apply --auto-approve

16. Verify the updated resource in the AWS portal ...

Security Groups (6) Info			
Find security groups by attribute or tag			
☐	Name ▾	Security group ID ▾	Security group name
☐	secgp-db	sg-05e1c3332390498f5	secgp-3
☐	secgp-app	sg-051766b8238f5703a	secgp-2
☐	secgp-web	sg-05e98f5b894dfed98	secgp-1
☐	secgp-admin	sg-093819adde60bd368	secgp-0

17. In **terraform.tfvars**, comment line 12 and uncomment line 13 so that **web** is removed from the list.

Before running Terraform, stop and consider what you expect to happen. The Name attribute for security groups [1] and [2] have changed and security group [3] is no longer needed. The mutable attribute changes will therefore triggers change actions for security groups [1] and [2] along with the destruction of security group [3].

18. Plan the changes:

terraform plan

Review the plan carefully, it should show 0 to add, 2 to change, 1 to destroy.

Terraform is still using count, so the resources are still addressed as [0], [1], [2] and so on. However, the values attached to those positions have shifted.

That means:

demo[0] (secgp-0) Name **secgp-admin** remains unchanged
 demo[1] (secgp-1) Name **secgp-web** changes to **secgp-app**
 demo[2] (secgp-2) Name **secgp-app** changes to **secgp-db**
 demo[3] (secgp-3) is destroyed

19. Apply the changes:

terraform apply --auto-approve

20. Verify the new resources in the AWS portal ...

Security Groups (5) Info			
Find security groups by attribute or tag			
☐	Name ▾	Security group ID ▾	Security group name
☐	secgp-db	sg-051766b8238f5703a	secgp-2
☐	secgp-app	sg-05e98f5b894dfed98	secgp-1
☐	secgp-admin	sg-093819adde60bd368	secgp-0

What this demonstrates: What looks like a small logical change can produce a much larger operational effect. A simple ordered list is fragile. Even when your intent is small, ordinal indexing can cause widespread change because resource configuration is tied to position.

21. In **terraform.tfvars**, comment line 13 and uncomment line 14 so the list is empty. This will remove all existing security groups ahead of the next phase.
22. Apply the changes:

terraform apply --auto-approve

Phase 3 – Identity churn due to immutable attribute change

23. In **main.tf**, comment line 43 and uncomment line 44. This binds the 'Security group name' to be given to the security group to the explicit positional value in the security_groups variables. A positional change will invoke a 'Security group name' change, an immutable attribute in AWS.
24. In **terraform.tfvars**, comment line 14 and uncomment line 10. This adds 2 elements to the security_groups list.

25. Plan the changes:

terraform plan

Review the state addresses again after apply:

terraform apply --auto-approve
terraform state list

You should still see addresses such as:

```
aws_security_group.demo[0]  
aws_security_group.demo[1]
```

terraform state show aws_security_group.demo[0]

```
name           = "secgp-web"  
name_prefix    = null  
owner_id       = "503561439433"  
revoke_rules_on_delete = false  
tags           = {  
  "Name" = "secgp-web"  
  "Role" = "web"  
}
```

26. Verify the new resources in the AWS portal ...

Security Groups (4) Info			
Find security groups by attribute or tag			
☐	Name	Security group ID	Security group name
☐	secgp-web	sg-08e6d327e252c460c	secgp-web
☐	secgp-app	sg-0f63a3dbc470fd0d0	secgp-app

27. In **terraform.tfvars**, comment line 10 and uncomment line 11 so that **db** is added at the end of the list.
28. Plan the changes:

terraform plan

The plan should show 1 to add, 0 to change, 0 to destroy.

The new resource will be aws_security_group.demo[2]. This will have no effect on the existing resources [0] and [1], because their positions in the list have not changed.

29. Apply the changes:

terraform apply --auto-approve

30. Verify the updated resources in the AWS portal ...

Security Groups (5) Info			
Find security groups by attribute or tag			
☐	Name	Security group ID	Security group name
☐	secgp-web	sg-08e6d327e252c460c	secgp-web
☐	secgp-db	sg-083c6e98a25a5b605	secgp-db
☐	secgp-app	sg-0f63a3dbc470fd0d0	secgp-app

31. As before, when using count, adding items to the end of a list usually produces stable behaviour. Terraform adds the new resource and leaves the existing indexed resources untouched.

32. In **terraform.tfvars**, comment line 11 and uncomment line 12 so that **admin** is inserted at the start of the list.

33. Plan the changes:

terraform plan

Review the plan carefully, it should show 4 to add, 0 to change, 3 to destroy.

```
Changes to Outputs:
~ security_group_addresses = [
  - "aws_security_group.demo[0] => secgp-web",
  - "aws_security_group.demo[1] => secgp-app",
  - "aws_security_group.demo[2] => secgp-db",
  + "aws_security_group.demo[0] => secgp-admin",
  + "aws_security_group.demo[1] => secgp-web",
  + "aws_security_group.demo[2] => secgp-app",
  + "aws_security_group.demo[3] => secgp-db",
]
```

The 'Security group name' used for the security groups [0],[1] and [2] has now changed. This is an immutable attribute and therefore triggers destroy/add actions along with the creation of security group [3].

34. Apply the changes:

terraform apply --parallelism=1 --auto-approve

35. Verify the **new** resources in the AWS portal ...

Security Groups (6) Info			
Find security groups by attribute or tag			
☐	Name	Security group ID	Security group name
☐	secgp-web	sg-01c8d42b2b28ce5dd	secgp-web
☐	secgp-db	sg-03b8ef5c988826820	secgp-db
☐	secgp-app	sg-049e8c48c51dd6a82	secgp-app
☐	secgp-admin	sg-04be5720185a18b2f	secgp-admin

Note the new IDs !

36. In **terraform.tfvars**, comment line 12 and uncomment line 13 so that **web** is removed from the list.

Before running Terraform, stop and consider what you expect to happen. The 'Security group name' used for the security groups [1] and [2] have changed and security group [3] is no longer needed. The immutable attribute changes will therefore triggers destroy/add actions for security groups [1] and [2] along with the destruction of security group [3].

37. Plan the changes:

terraform plan

Review the plan carefully, it should show 2 to add, 0 to change, 3 to destroy

```
Changes to Outputs:
~ security_group_addresses = [
  "aws_security_group.demo[0] => secgp-admin",
- "aws_security_group.demo[1] => secgp-web",
- "aws_security_group.demo[2] => secgp-app",
- "aws_security_group.demo[3] => secgp-db",
+ "aws_security_group.demo[1] => secgp-app",
+ "aws_security_group.demo[2] => secgp-db",
]
```

38. Verify the **new** resources in the AWS portal ...

Security Groups (5) Info				
Find security groups by attribute or tag				
☐	Name	Security group ID	Security group name	
☐	secgp-db	sg-09b3726227f9a3a36	secgp-db	
☐	secgp-app	sg-05b487ba2db7b6ddd	secgp-app	
☐	secgp-admin	sg-04be5720185a18b2f	secgp-admin	

Note the new IDs for secgp-db and secgp-app !

39. In **terraform.tfvars**, comment line 13 and uncomment line 14 so the list is empty.
This will remove all existing security groups ahead of the next phase.

40. Apply the changes:

terraform apply --auto-approve

Phase 4 – Stable keys with for_each

What Is Changing Here?

Up to this point, resources have been created using count, which means Terraform identifies them by **numeric index** (e.g. [0], [1]). These indexes are tied to the **position of items in a list**, not what those items represent. This is why reordering or inserting values earlier in the list caused widespread and often unexpected changes.

In this phase, we change two things:

- The input variable moves from a **list** to a **map**
- The resource moves from using count to using for_each

This introduces a fundamental shift in how Terraform tracks resources.

Terraform does not understand what a resource *means* — only how it is *addressed*

Instead of:

- aws_security_group.demo[0]
- aws_security_group.demo[1]

Terraform will now use:

- `aws_security_group.demo["web"]`
- `aws_security_group.demo["app"]`

Resources will no longer be identified by position, but by **stable keys**.

This means:

- Reordering input data no longer causes changes
- Adding new entries does not affect existing resources
- Terraform can track resources based on logical identity rather than list position

This is the key step that removes the instability seen in earlier phases and leads to a more predictable and maintainable design

41. In **variables.tf**, comment line 27 and uncomment lines 30. This re-types the variable `security_groups` from `list(string)` to `map(string)`.
42. In **main.tf**, uncomment lines 39 and 55, and comment lines 57 and 73. This switches to a resource block that references the re-typed `security_groups` variables.
43. In **terraform.tfvars**, comment line 14 and uncomment line 17. This populates the re-typed `security_groups` variable with initial values.
44. Before running Terraform, consider what is about to change. The resources will no longer be addressed by numeric index.
45. Plan the changes:

terraform plan

Previously, Terraform used addresses such as:

```
aws_security_group.demo[0]
aws_security_group.demo[1]
```

The new resource block in `main.tf` uses `for_each` with a map, so Terraform will now use key-based addresses such as:

```
aws_security_group.demo["web"]
aws_security_group.demo["app"]
```

This is the critical shift: Terraform is no longer identifying resources by their position in a list, but by a stable, explicit key.

46. Apply the changes:

terraform apply --auto-approve

47. Verify the new state addresses:

terraform state list

You should now see key-based addresses rather than numeric indexes.

```
aws_default_security_group.default
aws_security_group.demo["app"]
aws_security_group.demo["web"]
aws_subnet.demo
aws_vpc.main
```

Pause and reflect: Nothing about the infrastructure itself has become more complex. What has changed is how Terraform identifies each resource. By introducing explicit keys, we have removed ambiguity from resource identity.

Proving the stable design

48. In **terraform.tfvars**, comment line 17 and uncomment line 18 so that **db** is inserted between **web** and **app**.

49. Plan the changes:

```
terraform plan
```

The plan should show 1 to add, 0 to change, 0 to destroy.

The new resource will be `aws_security_group.demo[db]`. This will have no effect on the existing resources `[web]` and `[app]`, because their keys are unchanged and are not tied to their positions in the list.

```
terraform apply --auto-approve
```

50. In **terraform.tfvars**, comment line 18 and uncomment lines 19. This re-orders the values for the `security_groups` variable.

51. Plan the changes:

```
terraform plan
```

The plan should show no changes.

Why? Because although the order of the entries in the list has changed, the keys have not. Terraform is using those keys for identity, so reordering the map does not cause churn.

This is the stable end state the lab is designed to lead you toward. The map provides explicit keys, and `for_each` uses those keys to maintain stable addressing in state. This pattern is known as **key-based resource addressing**.

Best-practice conclusion: For long-lived infrastructure, prefer `for_each` with stable map keys and explicit structured values. Use simple ordered lists with care, because ordinal indexing is fragile.

Phase 5 – Clean up

52. Destroy everything created in this lab:

```
terraform destroy --auto-approve
```

Transition to Lab 2 – From Stable Identity to Rich Data

In this lab, the focus has been on understanding how Terraform tracks resources and how input data structures influence stability over time.

To keep that focus clear, we deliberately used a simple data structure:

- `map(string)`

This allowed us to concentrate on **resource identity and churn**, without introducing additional complexity.

In real-world configurations, however, a single string is rarely sufficient to describe a resource. Infrastructure definitions typically require multiple attributes, such as naming conventions, environment context, tagging strategies, and configuration options.

To support this, Terraform commonly uses more expressive data structures such as:

- `map(object)`

This allows each resource to be defined with a richer, structured set of values while still benefiting from the **stable key-based identity model introduced by `for_each`**.

However, introducing more complex input structures also introduces new challenges:

- Input data may be incomplete, inconsistent, or poorly formatted
- Values may need to be normalised, validated, or transformed before use
- Small inconsistencies can propagate into configuration errors or unexpected behaviour

What comes next

In Lab 2, we build on the stable design established here and introduce **data cleansing and transformation techniques**.

You will:

- Transition from `map(string)` to `map(object)`
- Work with more realistic, structured input data
- Learn how to normalise and validate inputs before they are used in resources
- Apply Terraform expressions and functions to transform data into a consistent, usable form

The focus now shifts from:

- **“How does Terraform track resources?”**

to:

- **“How do we prepare and control the data that drives those resources?”**

****** Congratulations, you have completed this lab. ******

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